

Methodology and validation

Sean Muller · October 2026

The standard a research result must pass before I believe it, and four cases where applying it cut a result of mine, withdrew one or found nothing to claim.

EVERY PERFORMANCE FIGURE IN THIS BRIEF IS A HISTORICAL BACKTEST. NO LIVE OR PAPER-TRADING RESULTS ARE SHOWN.

3.04×

Historical backtest ·
selected configuration:
ending multiple

0.85

Its deflated Sharpe ratio,
against 240 simulations

1 of 69

Candidates surviving one
family-wise test: already
known, fragile

540

Runs found by the trial
recount, approximately;
100 first assumed

Selected configuration result. Configuration selection and evaluation overlap in this research window, so it is not independent out-of-sample evidence. Deflated Sharpe ratio: 0.85.

The subject is a systematic U.S. equity strategy that trades relative-value effects among single stocks, under portfolio-level risk controls. This brief documents one configuration of that strategy, which is long-only, and covers how claims about it are tested. How it works is not described.

The standard

I believe a result only after it passes six steps.

1. **Pre-register.** Hypothesis, baseline, metric, placebo and stop rule are written before the result is read.
2. **Baseline.** A variant is judged against the standing baseline, not against zero.
3. **Placebo.** A copy with the same mechanics and its information removed must fail, or the claim is withdrawn.
4. **Era split.** Choices are set on the earlier half of the sample and judged on the later half.
5. **Same-build replay.** Variant and control run on one pinned software build and the same data.
6. **Family-wise check.** All leads from a sweep are tested together, with placebo twins in the panel.

The October 2026 research sweep ran under this standard; the audit and trial recount below followed a written protocol without per-study pre-registration. The research questions, methodology, system design, and interpretation are independently directed; implementation uses extensive AI-assisted development. The headline backtest is a selected result rather than a product of this standard: its liquidity requirement was chosen on the full window, and the revised screen did not beat its placebos (Cases 1 and 4).

Case 1: auditing my own first reported backtest

In late September 2026 I reported a two-year walk-forward backtest: models fitted on past data, then applied to later data they had not seen. In early October I had it audited point-in-time: every input checked for whether it was knowable on the date it was used.

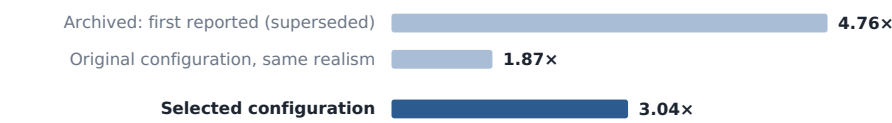
The list of tradable stocks was not: one 2026 snapshot had been applied to every earlier date. Orders also filled at the signal price with no delay, and stocks that delisted during the window were missing, as they still are. Walk-forward testing had protected the fitted parameters, not the universe.

I had the backtest rebuilt: first the universe as of each date under a point-in-time listing and minimum-liquidity rule, then with execution costs and fill uncertainty modeled as functions of spread, liquidity, volatility and order size rather than assuming full fills at the signal price. The flat fee is kept inside the model. Each change reduced the result materially. I accepted the lower figure because the evidence behind it is more credible.

BACKTEST	ENDING MULTIPLE	SHARPE	MAX DRAWDOWN	TRADES
Archived · historical backtest, first reported, superseded	4.76×	2.32	−16.6%	not shown
Historical backtest · original configuration, same realism (point-in-time universe, modeled fills)	1.87×	1.25	−17.4%	583
Historical backtest · selected configuration	3.04×	1.84	−19.1%	780

Historical backtests of one long-only account, \$1M start, profits reinvested, Jun 2024 – Jul 2026. The first row, superseded, ends Jun 2026. Sharpe on daily returns, annualized, no risk-free rate subtracted.

Selected configuration result. Configuration selection and evaluation overlap in this research window, so it is not independent out-of-sample evidence. Deflated Sharpe ratio: 0.85.



Ending value of \$1 after about two years, in three historical backtests: first reported (archived, superseded), the original configuration at the same realism, and the selected configuration. Bars are drawn from 1×, the starting dollar. The first bar’s window ends about a week before the others. The last bar is not a further audit step: its liquidity requirement was chosen afterward, on this same window.

What changed. Breadth: the trade count fell sharply once the universe was rebuilt as of each date.

Lost breadth raised the next question: whether the liquidity rule was set well. The selected configuration keeps the point-in-time universe and modeled fills, and revises how the liquidity screen is measured as of each date, and the level of the minimum-liquidity requirement. The second revision restores breadth.

I chose the requirement after testing five settings on this same window. Every revised setting beat the original after costs, and several settings are within noise of one another. A two-year compounding multiple moves materially under small, arbitrary changes to a rule, so differences of that size between runs are not read as skill.

Case 2: one family-wise test over a whole sweep

The October 2026 sweep counted more than 10,000 configurations on the same two years, so some would pass by luck. The sweep took the first reported backtest (superseded: hindsight universe, flat cost) as its reference; a second sweep on the selected configuration is on the research agenda. Its 69 candidate improvements with a per-session return series therefore entered one family-wise test, which controls the chance of even one false discovery: Reality Check and SPA with a Romano–Wolf step-down, on a block bootstrap over sessions. Forty-four placebo twins, copies of the selection-type candidates with the information removed, formed a control family.

One candidate survived at 5%. It was an effect the program already knew, and it is fragile: its adjusted p-value is 0.005 at the study's primary block length, 0.04 at the next, and 0.14 and 0.21 at the two longest. On the later half of the sample alone it never clears 5%. One different candidate does, at the longest block only, which the study's own rule classes as unstable.

The twins behaved as a null should (family-wise p-value 0.91, no rejections), and I claimed no new lever. Planted effects of the two sizes tested were recovered about a fifth and about a third of the time at the studies' own thresholds, so most nulls here mean below resolution, not absent.

Case 3: measuring the noise ceiling before modeling

Several studies had tried to rank individual signals by quality and found nothing. Before another model was fitted, a feasibility study measured the noise ceiling: the score a ranker would reach if it knew every signal's true expected value.

The ceiling is low. Noise in one signal's outcome is more than ten times the true spread in expected value between signals, so a perfect ranker would score barely above chance, where every earlier attempt had landed. Under its pre-registered feasibility rule, that line of modeling was closed as below what this sample can resolve.

Case 4: a claim withdrawn when its placebo matched it

A revision to how the liquidity screen is measured looked strong at first: with modeled costs it ended materially above the original configuration, and the research log called it the rule to bring forward. A formal test then put its advantage at about 1.4 standard errors (family-wise adjusted p-value about 0.35), and the log entry was corrected in writing as premature.

Then the placebos ran: the same mechanics with the information replaced by a random draw. With modeled costs, the placebos mostly finished level with or above the real version. Before costs it sat below its like-for-like control and inside the range of three placebos.

The claim was withdrawn, and I credit the revised screen with no skill of its own. The selected configuration still measures liquidity this way: the requirement settings were tested on that screen. So that revision carries no claim of skill: the gap to the best placebo is inside the noise band of a two-year multiple.

Trial accounting

A deflated Sharpe ratio places the observed Sharpe on a probability scale against the best that luck alone would be expected to produce across the trials run, so it is only as good as its trial count. My first reported backtest was deflated for an assumed 100 trials. I had the record recounted: about 540 simulation runs in 62 hypothesis families over the same two-year period.

The count is a range. Surviving daily return series correlate 0.88 on average, so near-duplicate runs are not independent trials: about 1 to about 30 by correlation, 540 by raw count. The first-reported Sharpe of 2.32 (superseded) survives a simple family-wise correction at 5% for up to 55 independent trials, so not at the 100 first assumed.

For the selected configuration the deflated Sharpe ratio is 0.85 against 240 full-window simulations on file, below the conventional bar of 0.95. Simulations on file are not ideas tried, so 240 may understate the search.

What is completed and what is next

COMPLETED	NEXT
Point-in-time universe rebuild (being completed) and modeled fills	Start-date sensitivity on the selected configuration; the first-reported start was the best of five tested
Liquidity requirement tested at five settings; placebos of the revised screen	Formal concentration, interval and factor-attribution studies on the selected configuration
One family-wise test over the sweep, with placebo twins Live execution evidence used to refine implementation-shortfall and impact estimates as the sample grows	
Trial recount and deflated Sharpe ratio for the selected configuration; a parity check that the research copy of the replay reproduces the reference backtest	A forward test, with the configuration fixed before later data is examined

Limits

- The headline is a historical backtest: one implementation of one two-year window.
- Fitted parameters are out of sample in a walk-forward test. The choice among configurations is not: it used the same window, so the result is not independent out-of-sample evidence.
- Profit is concentrated. A small fraction of trades accounts for more than half of net profit, and the final mark is about 18% below the peak.
- Pre-registration is internal and self-dated. For 16 of the 18 sweep studies the entry preceded the report in version control. The sweep-wide family-wise study is one of the two exceptions. Version control dates each entry before its report, not before the work began: eight of the sixteen were first committed already marked as running. One placebo panel was written after the direction of its comparison was known.
- The work has not yet had independent outside review. One earlier look-ahead audit shared a defect with the code it checked, and passed it.

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